Lesson notes · for you

The shortest way

Optimization · Level 1 · Lesson 2

Time	Materials
20 min	Paper and a pencil; The map from the lesson

Learning goals

- Add up road lengths to find how long a way is.
- Use the closest-first trick to find the shortest way.
- Find a new way when a road is closed.

Activity: Map detectives

1. Guess (3 min): Look at the lesson's map and write down which way you think is shortest.
2. Add (5 min): Add up the way through Bora (5 + 10) and the way through Tala, Kesi and Mina (4 + 3 + 2 + 4). Was your guess right?
3. Trick (8 min): Write 0 next to Home. Each turn, circle the waiting village with the smallest number, then write new numbers on its neighbours. Stop when the school is circled.
4. Flood (4 min): Cross out the bridge between Kesi and Mina and do the trick again. What is the new shortest way?

Copy the map from the lesson onto paper. No device needed.

Discussion

- Why did the straight-looking way lose?
- What changed when the bridge flooded?
- Who walks the farthest to your school?

Check yourself

- The shortest way is ___ km. With the bridge closed, it is ___ km.

Try it this week

Draw a map of your way to school or to the market, with a number on each road (how many minutes it takes to walk). Find the shortest way with the closest-first trick. Then pretend one road is closed and find the new way.

Worksheet

The shortest way

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Name _____ Class _____ Date _____

1 **After heavy rain, the bridge between Kesi and Mina is closed. Which way from home to school is the shortest now?**

◯ Through Tala and Kesi: 15 km ◯ Through Rubi and Sana: 14 km ◯ Through Bora: 15 km

2 **The road that looks the straightest is always the shortest way.**

◯ True ◯ False

3 **Put the closest-first trick in order.**

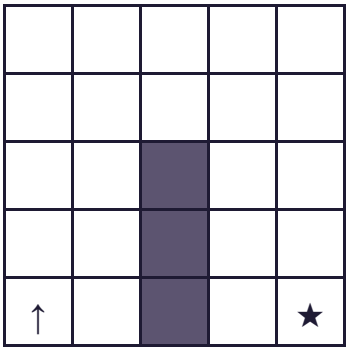
☐ Settle the waiting village with the smallest number

☐ Give home the number 0

☐ Keep going until the school is settled

☐ Write smaller numbers on its neighbours if you can

4 **A wall is in the way! Take the robot to the star by the shortest way. Use 6 cards or fewer.**



Cut out cards (↑ forward, ↶ turn left, ↷ turn right) and lay out a program that takes the robot to the ★. Write it here:

_____

_____

5 **How long is the way Home → Tala → Kesi → Mina → School? The roads are 4, 3, 2 and 4 km.**

◯ 15 km ◯ 13 km ◯ 4 km

✎ **Think of your way to school. Is it the shortest way, the safest, or the one you like best? Why do you go that way?**

_____

_____

_____

Check your answers

The shortest way

1 **Through Rubi and Sana: 14 km**

With the bridge closed, add up the ways that are left: through Rubi and Sana, 6 + 5 + 3 = 14 km.

2 **False**

You have to add up the roads. Through Bora looks straight but is 15 km; the bendy way through Tala, Kesi and Mina is 13 km.

3 **Top to bottom: 2, 1, 4, 3**

Start at home with 0, then repeat: settle the smallest number, and update its neighbours.

4 **Any program that takes the robot to the ★ with 6 cards or fewer.**

Up three squares, turn right, along four, turn right, down three: ten moves, the fewest that get around the wall.

5 **13 km**

Add up all the roads on the way: 4 + 3 + 2 + 4 = 13 km.